

INKR: Bridging the gap between comic fans and creators with BigQuery and Vision AI

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INKR uses BigQuery to better understand user profiles and Vision AI to make comics more accessible for creators and fans around the world.



About INKR

INKR is a digital comic platform on a mission to make comics universally accessible through localization. It works with more than 100 publishers and comic creators from Japan, China, Korea, and the US to bring their content to the world. The company has 40 employees with offices in Singapore and Vietnam and more than 70,000 monthly active users worldwide.

Google
Cloud
results

Imp
to p

Industries: Media & Entertainment

- Personalizes content recommendation to 20 million+ monthly active users
- Reduces model training cost by 60% with Compute Engine automation
- Automates the transcription process with Vision AI for localization
- Creates professional business reports for partners easily with Looker Studio

from
week
day

Location: Singapore

Data Studio

Member of the InfoFabrica Group



About Cloud Kinetics

With a mission to transform customers' business to realize the power of cloud, Cloud Kinetics is a Google Cloud Premier Partner that helps enterprise businesses across industries accelerate their cloud initiatives through a wide range of solutions, with a customer-first approach.

With a mission to transform customers'

Google Cloud (<https://cloud.google.com/>) d,

Cloud Kinetics is a Google Cloud Premier

BigQuery (<https://cloud.google.com/bigquery/>)

Partner that helps enterprise businesses

Compute Engine

(<https://cloud.google.com/compute/>)

Looker Studio

(<https://marketingplatform.google.com/about/data-studio/>)

Comic readers

Vision AI (<https://cloud.google.com/vision/>)

Imagine seeing a comic cover that you want to read but being unable to read and enjoy it because it's in a foreign language. Comic readers may know this dilemma all too well. The reason that [INKR](https://inkr.com/) (<https://inkr.com/>) was founded back in 2019 with a mission to make all comics more universally accessible. "There are so many amazing comics from countries like Japan, Korea, and China that don't get the visibility they should because of language barriers," says Hieu Tran, co-founder and Head of Operations at INKR. By translating these

comics into English and publishing them online, comic creators and readers can have wider access to a range of different titles.

Today, the team works with more than 100 publishers and creators from around the world to give visibility to their content. As a digital platform, INKR is always looking for ways it can use technology to help publishers and creators to localize efficiently and publish directly to their readers. It also gives comic fans access to a library of content in their local language on the INKR platform.

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—Vu Hoang, Head of Data
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“The traditional way of licensing can be pretty cumbersome and time consuming, not to mention a very manual process. This is where we come in by providing the tools and technologies to help creators reach readers in a more efficient way,” says Tran.

INKR has been using Google Cloud (<https://cloud.google.com/>) for six years and works with Cloud Kinetics (<https://www.cloud-kinetics.com/>) for administrative support. “Before Google Cloud, we did not have a tool to properly collect and analyze data,” says Vu Hoang, Head of Data and AI at INKR. It was only when the team started to notice a growth in user base that they decided to start looking into

different cloud platforms for data collection and analysis. That was when they discovered Google Cloud. “We chose Google Cloud because of its scalability and the fact that it has no infrastructure and maintenance cost,” explains Hoang. “As a startup, cost is an important factor, and not having to allocate cost for maintenance was a huge benefit, not to mention the integration with Firebase Analytics, which is what we use for mobile development.”

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—Vu Hoang, Head of Data
and AI, INKR

Giving comic fans the content they want by leveraging BigQuery and Compute Engine

To get a clearer understanding of who its audiences were, the first product INKR explored with Google Cloud was BigQuery (/bigquery). With more than 70,000 monthly active users globally, INKR uses Firebase to ingest user data into BigQuery as a centralized data warehouse. It also ingests raw data from backend systems into Cloud Storage (/storage) and runs an ETL pipeline into BigQuery.

“BigQuery is ultimately our centralized data warehouse that’s responsible for all our data. We also use BigQuery to create business reports for partners and internal use, and training data for our machine learning (ML) models like our recommendation system,” says Hoang. With data from BigQuery, INKR has a better understanding of the titles and genres that fans enjoy and can be more focused in sourcing content for its platform.

Where reporting is concerned, the INKR team uses

Looker Studio

(<https://marketingplatform.google.com/about/data-studio/>)

to produce weekly reports that help internal stakeholders and teams make better, data-backed decisions. “Looker Studio is a very professional looking platform that’s quick and easy to use. This makes reporting tasks a breeze,” explains Hoang. Because Looker Studio can be directly integrated with BigQuery, the team doesn’t have to manually extract data to be presented. Instead, they can simply pick and choose the information they want, and Looker Studio would lay it out automatically in the most presentable way.

The recommendation model on INKR is also trained regularly to cater to each member’s profile and their interests. While the company used to run the model training process on-premises, the flexibility for customization on [Compute Engine](#) (/compute) has led INKR to fully run their models on Compute Engine. “Since we’ve moved the process to Compute Engine, the time to train a model has been reduced from several hours to less than an hour, with our time to production reduced to just one day compared to one week,” explains Hoang. The automation capabilities provided by Compute Engine have also freed up hours of engineers’ time to work on other projects.

Beyond saving time, Compute Engine has reduced the cost of model training by 60%, from \$100 a month to just \$40 a month, since there’s no infrastructure or maintenance cost attached to Compute Engine. “In the past, just transferring the data from BigQuery to our local network would have cost us almost \$100 a month before maintenance,” says Hoang. This reduction in operational cost

means that more budget can be allocated to other areas of the business, such as sourcing for new talent.



Comic localization with AI

Localization for comic books is an extremely laborious process, so having the right technologies to do the heavy lifting is crucial. “One of the most time consuming parts of this process is analyzing the images to generate metadata,” explains Hoang. From facial expressions of the characters to text bubbles and colours, INKR uses Cloud Vision (/vision) to analyze the images and generate the metadata. It also does text transcription automatically, which is a huge time saver for the team.

“The text transcription has really saved us hours of time because in the past, someone would have to read the content and type it out onto spreadsheets manually before it is passed to the translator,” explains Tran. The translated copy would then have

to be matched to each speech bubble manually. Vision AI can detect anything from faces to objects, even read handwritten copy, so translators can go directly onto the comics to type in the appropriate translation in an auto-sized typeset, where they can see the final output on how it would appear in the speech bubble.

“We now rely on Vision AI to automate the localization process as much as possible because it saves time and allows us to publish quality localized content much more quickly and efficiently,” says Tran.

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*—Hieu Tran, co-founder
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Exploring new horizons with more robust localization efforts using AI

Continuing on its mission to make comics more universally accessible through localization, INKR’s short-term goal is to expand into the European market, covering French, German, and Italian. In the longer term, the company will invest more in AI technologies to improve the localization process further. “This will require natural language processing technology and model training, which will require powerful infrastructures like Compute Engine for support,” says Hoang.

Tran adds, “We have different groups of creators and publishers and different groups of readers all around the world. Without the help of big data and AI enabled by Google Cloud, I don’t think we would have been able to build this platform we have today.”

**Manga, webtoons,
manhua, comics
in one app**

